TaskFlow Pro

Engineering a Production-Ready Application with Zero Dependencies

Built for:
Web Development Internship
Final Task (InternNova)

Author:
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Tech Stack:
Pure HTML5, CSS3, ES6+
JavaScript (No Frameworks)

The Zero-Dependency Layer Cake

Vanilla ES6+ JS

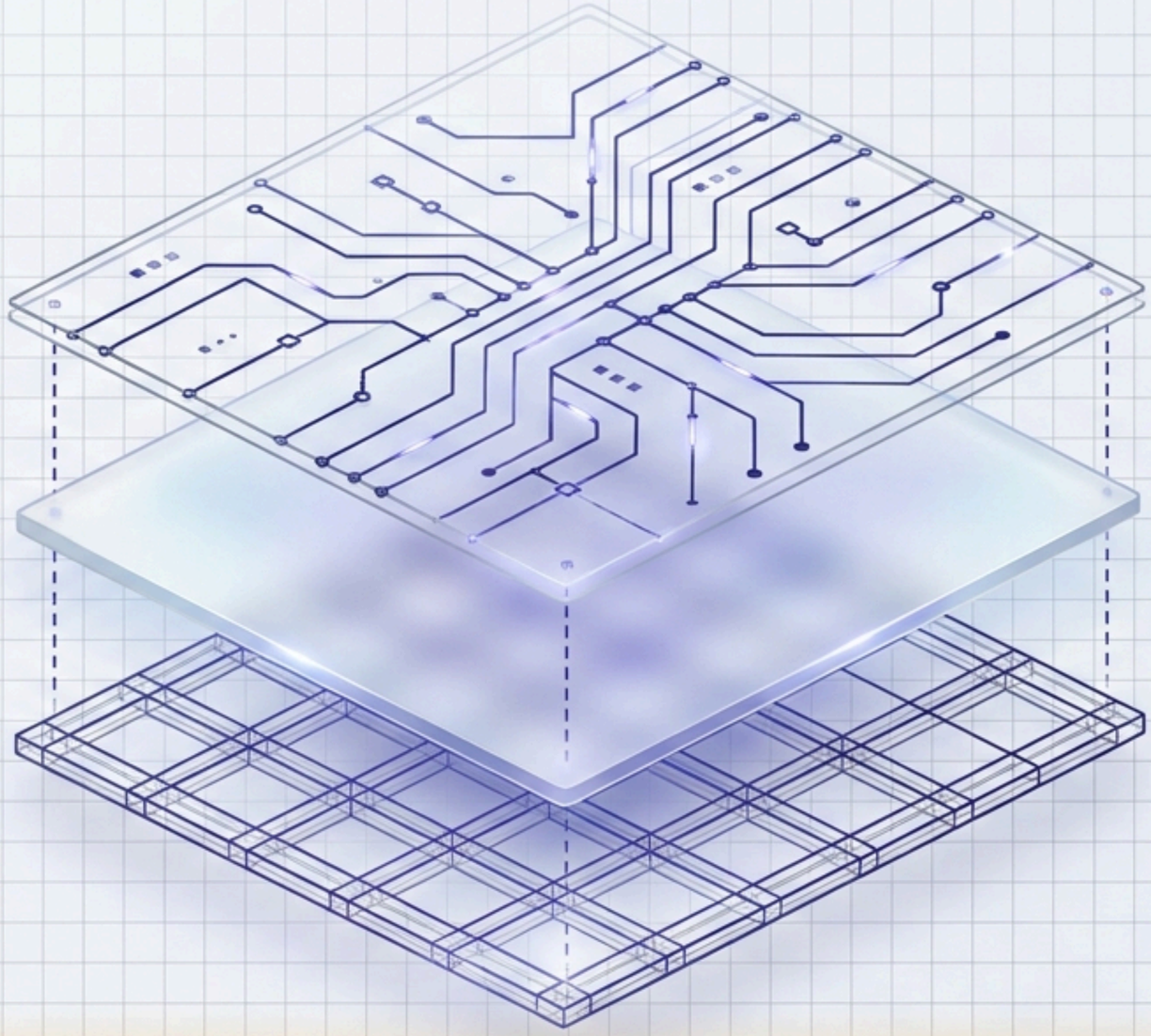
Real-time state management,
LocalStorage sync,
Filter/Sort engine.

Modern CSS3

Custom Properties,
Fluid Layouts,
Glassmorphism UI.

Semantic HTML5

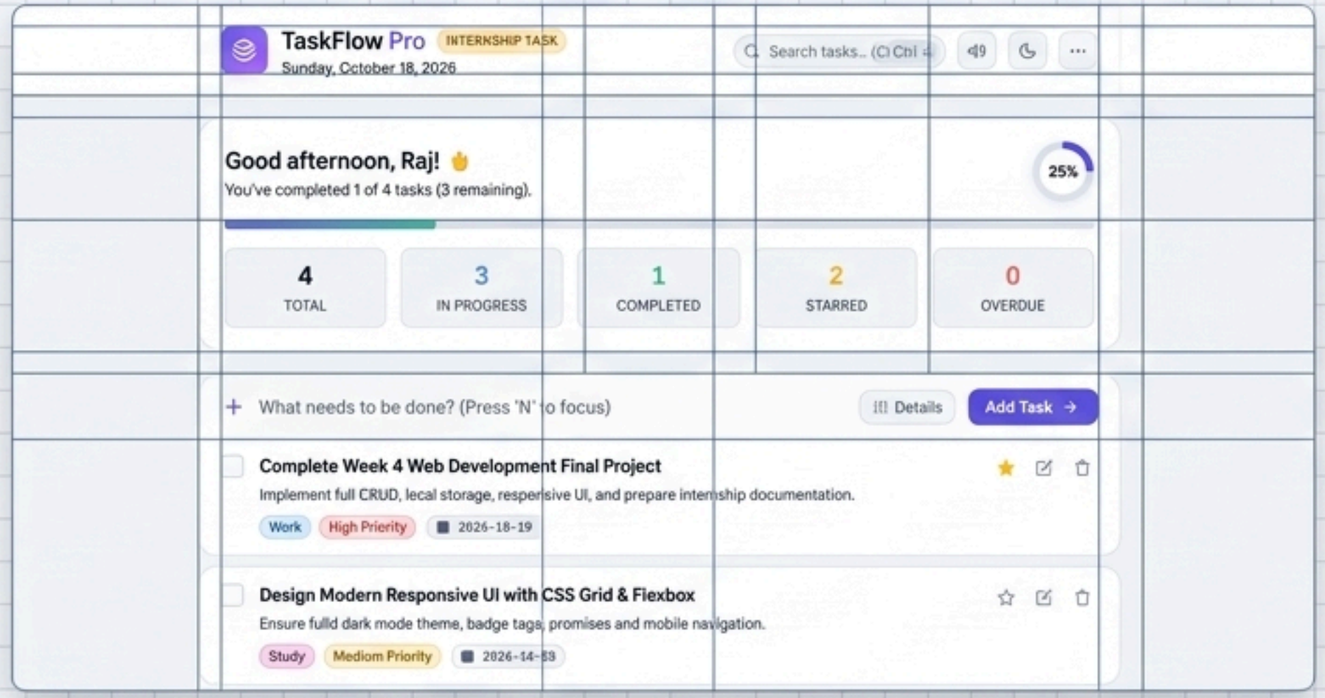
Accessible modal structure,
strict structural hierarchy.



No build tools. No external libraries. Runs natively out of the box.

The Responsive Blueprint

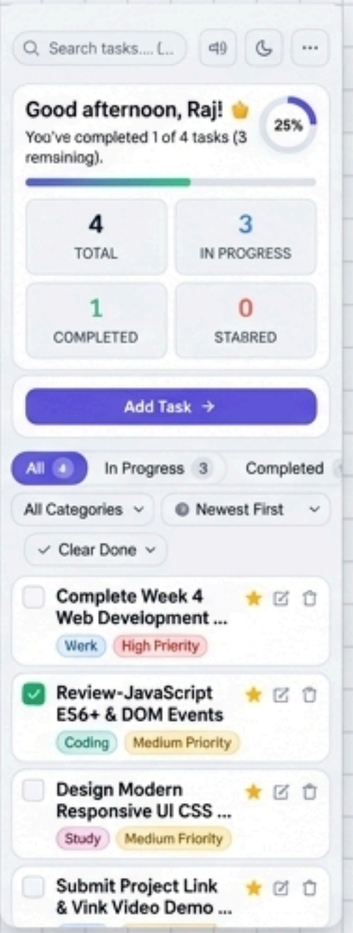
Desktop Context



Flexbox



Mobile Context



CSS Custom Properties dictate seamless Dark/Light theme switching.

Fluid typography and grid templates negate the need for JS-based layout intervention.

The Interaction Loop

User Intent

User presses “N” to focus, types task, presses Enter.



JS State Engine

Raw data parsed into JS Object (ID, Category, Priority, Timestamp).



State Committal

Array updated. Sent to LocalStorage via JSON.stringify.



DOM Re-Render

UI cleared and rebuilt from the updated State Array. Dynamic category badges injected.



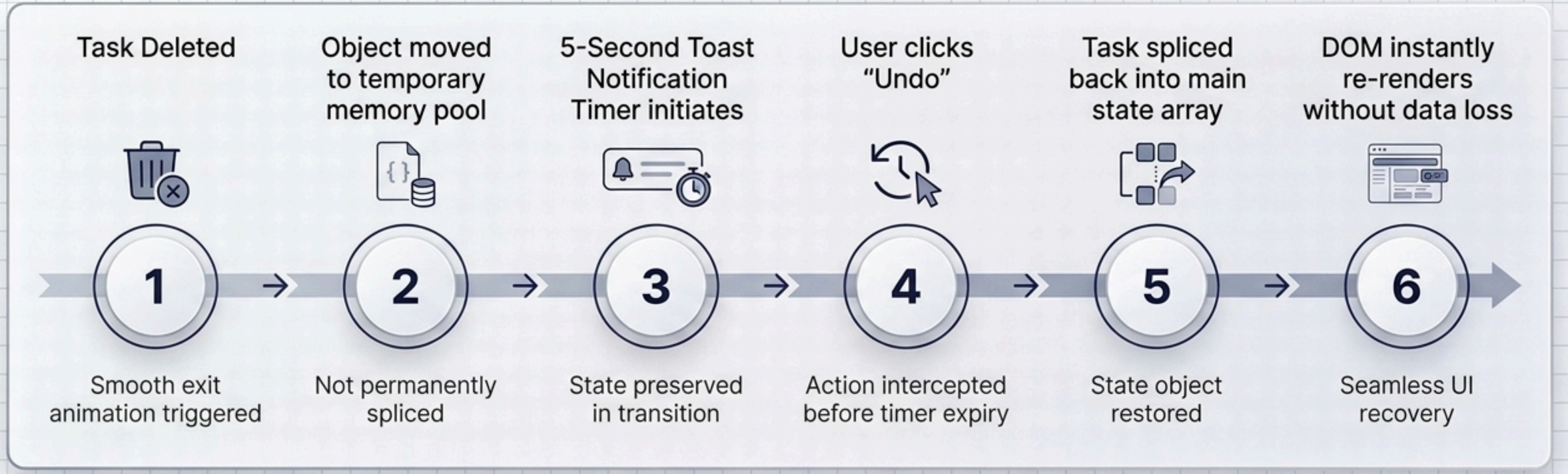
DOM Manipulation is strictly a consequence of State changes, ensuring perfectly synchronized views.

Multidimensional Querying Engine

User Intent	Status Filter	Category Filter	Sort / Search
Find urgent internship work.	In Progress	Work & Internship	Highest Priority
Review completed tech tasks.	Completed	Coding & Tech	Newest First
Locate specific upcoming deadline.	—	—	Search: ‘Ctrl + K’ Live Query Sort: Due Date (Earliest)

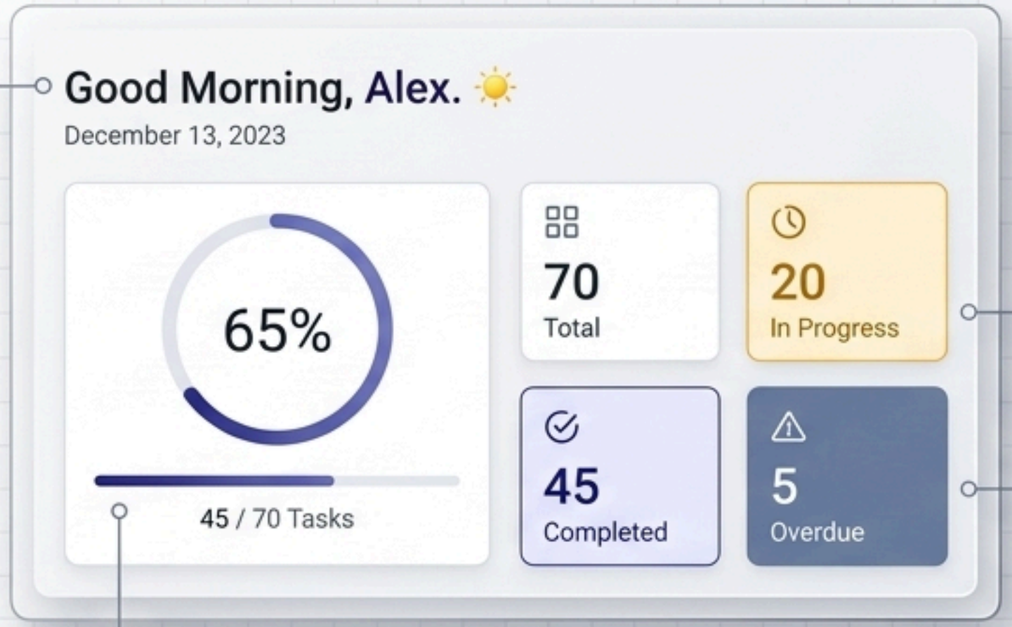
Real-time Live Search matches across task titles, descriptions, and categories simultaneously.

The ‘Undo Delete’ Micro-Timeline



> “Forgiving UX logic prioritizes data safety without sacrificing interface speed.”

Productivity Dashboard Anatomy



1

Dynamic Header

Calculates local time to dynamically inject personalized greeting.

2

Live Progress Badge

Progress derived in real-time by calculating (Completed Tasks / Total Tasks) * 100.

3

Interactive Counters

Counters serve as rapid-filter triggers. Clicking instantly updates the active view.

Sensory UX & Persistent Data

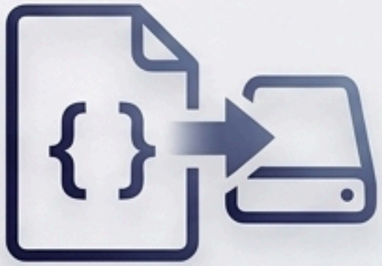
Sensory Feedback



Web Audio API Implementation

Provides pleasant, synthetic audio tones triggered on adding, completing, or deleting tasks. Fully toggleable by the user.

Data Portability

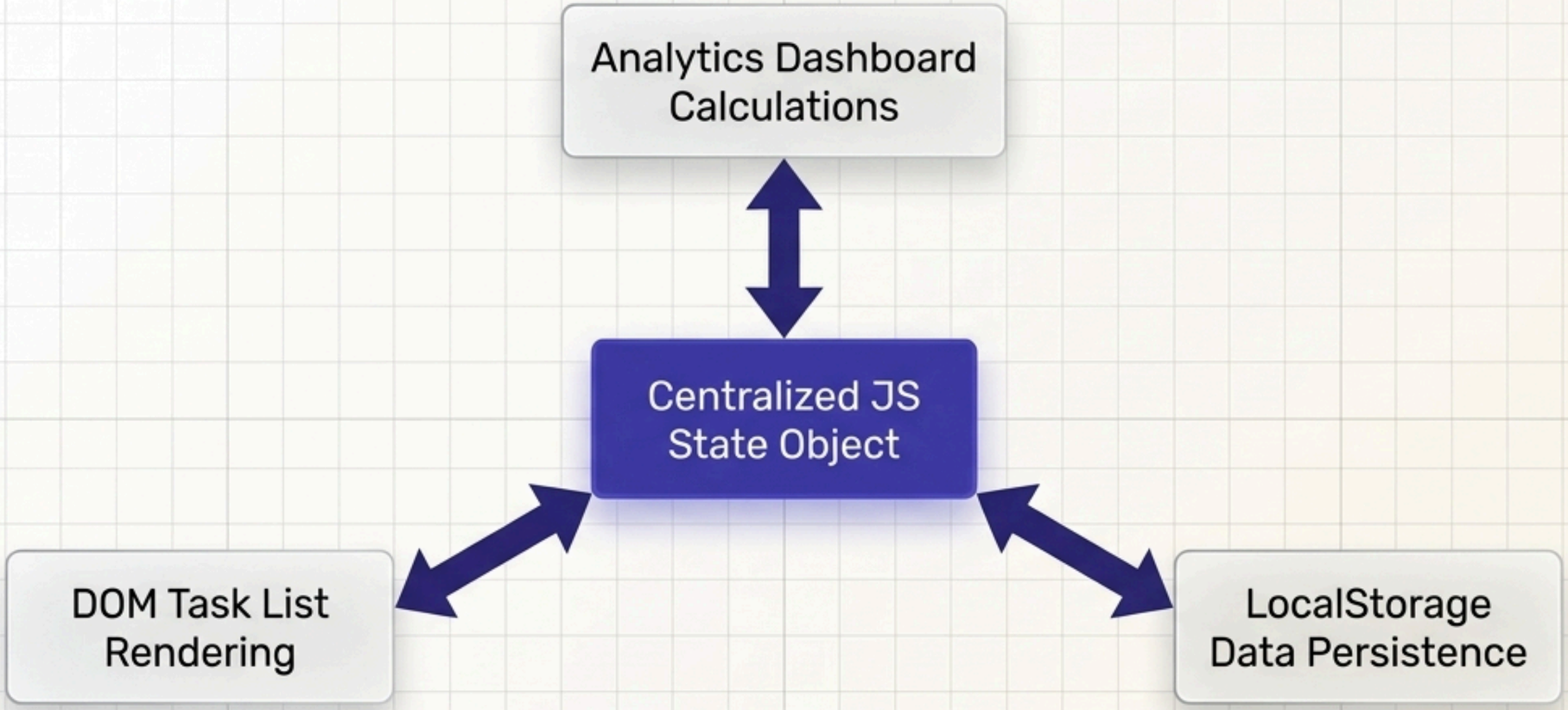


One-Click Data Backup & Restore

Allows users to export the localStorage state to a downloadable JSON file and import it back seamlessly to any browser.

Elevating utility to a premium user experience.

The Single Source of Truth



By decoupling the data from the DOM, TaskFlow Pro achieves complete synchronization. The UI never reads from itself; it strictly observes and reflects the central JavaScript array. This prevents state collisions and guarantees zero data drift across active sessions.

Launch Architecture & Deployment

Project Structure



Strict adherence to required files.
Zero bloat.

Deployment Environments

- GitHub Pages
- VS Code Live Server
- Local Python Server

Completely platform-agnostic. Requires no Node.js or server installation. Ready for instant evaluation.



THANK YOU

